

微視的テアリング不安定性にお けるリコネクション電場の導出 と検証

2021 年 6 月 8 日

兵庫県立大学大学院シミュレーション学研究科

FD18P004 柳生 光義



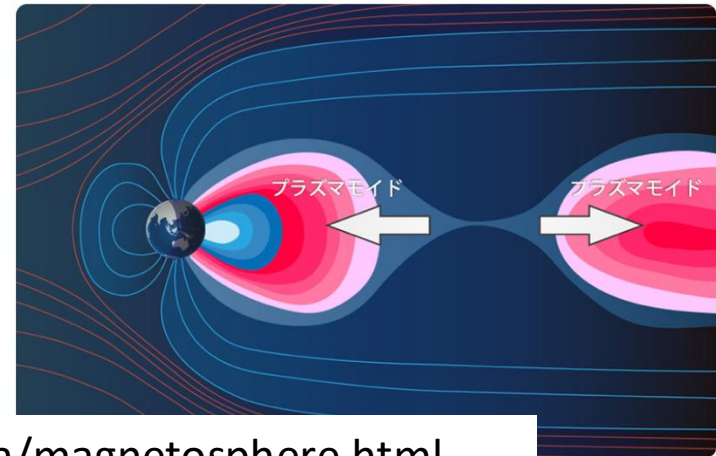
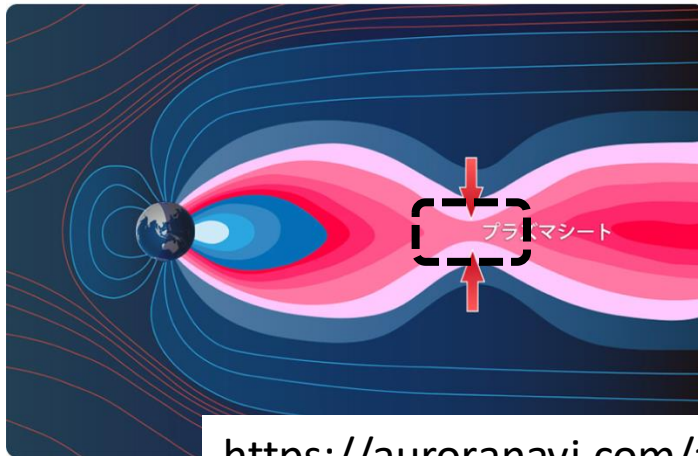
UNIVERSITY OF HYOGO
GRADUATE SCHOOL OF SIMULATION STUDIES

Microtearing mode (MTM)

- 電子圧力（温度）勾配が存在するとき，通常のテアリングモードが安定でも不安定になるモード（Hazeltine+, 1975）．
- 衝突性モード．無衝突の極限では安定．不安定化にはエネルギー依存する電子－イオン衝突が必要とされている（Drake+, 1977, Gladd+, 1980）．
- 複雑な磁場形状による粒子補足などによって，無衝突MT不安定性が存在する（Chowdhury+, 2016）．

Magnetic reconnection

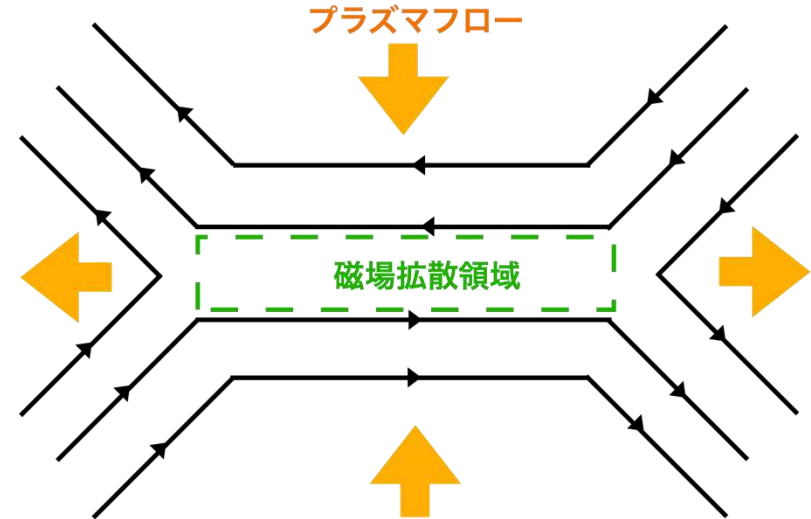
- ❑ MT不安定性は磁気リコネクションを伴う (Furth+, 1963).
- ❑ プラズマ中における普遍的な現象.
- ❑ 磁場のトポロジーを変え、磁場エネルギーを運動・熱エネルギーに変換する.



<https://auroranavi.com/aurora/magnetosphere.html>

Reconnection electric field

- ❑ 磁気リコネクション駆動には，磁場拡散領域においてリコネクション面の直交方向への電場が形成される必要がある．
- ❑ 電場はオームの法則によって求まる．
- ❑ オームの法則
via
電子の運動方程式



電子の運動論方程式より，電子の運動論効果を内包したオームの法則を導出し，MTMにおけるリコネクション電場を計算．

リコネクション電場の形成機構がMT不安定性に与える影響の調査は未実施．

- 粒子のジャイロ運動周期で平均化された運動論モデル.
- サイクロトロン周波数よりも低周波数の現象に着目 ($\Omega \gg \omega$).
- プラズマの粒子としての性質を捉える. 電子慣性やイオン有限ラーマ半径効果などを内包.

ジャイロ運動論オーダーリング

$$\frac{\omega}{\Omega} \sim \frac{k_{\parallel}}{k_{\perp}} \sim \epsilon$$

Electron kinetic eq.

電子分布関数

$$\delta f_e(\mathbf{r}, \mathbf{v}, t) = -\frac{q_e \phi}{T_{0e}} f_{0e} + h_e(\mathbf{R}, v_\perp, v_\parallel, t)$$

$$h_e(\mathbf{R}, v_\perp, v_\parallel, t) \quad : \text{リング分布関数} \quad -\frac{q_e \phi}{T_{0e}} \quad : \text{Boltzmann応答}$$

$$f_{0e}(x) = \frac{n_{0e}(x)}{(\sqrt{\pi} v_{\text{the}}(x))^3} e^{-v^2/v_{\text{the}}^2(x)} \quad : \text{平衡Maxwellian}$$

$$\text{電子運動論方程式} \quad \left(\frac{\partial h_e}{\partial t} \right)_c : \text{衝突作用素}$$

$$\begin{aligned} \frac{\partial h_e}{\partial t} + \mathbf{v}_E \cdot \nabla h_e + v_\parallel \hat{\mathbf{b}} \cdot \nabla h_e &= q_e \frac{f_{0e}}{T_{0e}} \frac{\partial}{\partial t} (\phi - v_\parallel A_\parallel) \\ &\quad - \frac{1}{B_0} \mathbf{e}_z \cdot \nabla (\phi - v_\parallel A_\parallel) \times \nabla f_{0e} + \left(\frac{\partial h_e}{\partial t} \right)_c \end{aligned}$$

Electron kinetic eq.

$$\begin{aligned} \frac{\partial h_e}{\partial t} + \mathbf{v}_E \cdot \nabla h_e + v_{\parallel} \hat{\mathbf{b}} \cdot \nabla h_e &= q_e \frac{f_{0e}}{T_{0e}} \frac{\partial}{\partial t} (\phi - v_{\parallel} A_{\parallel}) \\ &\quad - \frac{1}{B_0} \boxed{\mathbf{e}_z \cdot \nabla (\phi - v_{\parallel} A_{\parallel}) \times \nabla f_{0e}} + \left(\frac{\partial h_e}{\partial t} \right)_c \end{aligned}$$

右辺第2項を展開すると

$$\begin{aligned} \mathbf{e}_z \cdot \nabla (\phi - v_{\parallel} A_{\parallel}) \times \nabla f_{0e} &= \mathbf{e}_z \cdot \nabla \chi \times \nabla f_{0e} \quad (\because \phi - v_{\parallel} A_{\parallel} = \chi) \\ &= \cancel{\frac{\partial \chi}{\partial x} \frac{\partial f_{0e}}{\partial y}} - \frac{\partial \chi}{\partial y} \frac{\partial f_{0e}}{\partial x} \quad (= \{\chi, f_{0e}\}) \\ &= -\frac{\partial}{\partial y} (\phi - v_{\parallel} A_{\parallel}) \frac{\partial f_{0e}}{\partial x} \end{aligned}$$

Electron kinetic eq.

ここで,

$$\begin{aligned}\frac{\partial f_{0e}}{\partial x} &= \frac{\partial}{\partial x} \left[\frac{n_{0e}(x)}{(\sqrt{\pi} v_{the}(x))^3} e^{-v^2/v_{the}^2(x)} \right] \\ &= \frac{(n_{0e}(x))'}{(\sqrt{\pi} v_{the}(x))^3} e^{-v^2/v_{the}^2(x)} - 3 \frac{(v_{the}(x))'}{v_{the}^4(x)} \frac{n_{0e}(x)}{(\sqrt{\pi})^3} e^{-v^2/v_{the}^2(x)} \\ &\quad + 2 \frac{v^2}{v_{the}^3(x)} (v_{the}(x))' \frac{n_{0e}(x)}{(\sqrt{\pi} v_{the}(x))^3} e^{-v^2/v_{the}^2(x)} \\ &= \left[\frac{(n_{0e})'}{n_{0e}} + \left(\frac{v^2}{v_{the}^2} - \frac{3}{2} \right) \frac{2(v_{the})'}{v_{the}} \right] f_{0e} \\ &= \left[\frac{(n_{0e})'}{n_{0e}} + \left(\frac{v^2}{v_{the}^2} - \frac{3}{2} \right) \frac{(T_{0e})'}{T_{0e}} \right] f_{0e} \quad \left(\because v_{the}(x) = \sqrt{\frac{2T_{0e}(x)}{m_e}} \right)\end{aligned}$$

Gyrocenter dist. func.

Zocco+,2014のOrderingによるリング
分布関数

$$h_e = \left(\frac{q_e \phi}{T_{0e}} + \frac{\delta n_e}{n_{0e}} + \frac{2v_{\parallel} u_{\parallel e}}{v_{the}^2} \right) f_{0e} + g_e, \quad \int (1, v_{\parallel}) g_e d\mathbf{v} = 0$$

0th-Moment :

$$\int h_e d\mathbf{v} = \left(\frac{q_e \phi}{T_{0e}} + \frac{\delta n_e}{n_{0e}} \right) n_{0e}$$

1st-Moment :

$$\int v_{\parallel} h_e d\mathbf{v} = n_{0e} u_{\parallel e}$$

2nd-Moment :

$$m_e \int v_{\parallel}^2 h_e d\mathbf{v} = n_{0e} q_e \phi + \delta n_e T_{0e}$$

$$\frac{\delta T_{\parallel e}}{T_{0e}} = \frac{1}{n_{0e}} \int d\mathbf{v} \frac{2v_{\parallel}^2}{v_{the}^2} g_e + m_e \int v_{\parallel}^2 g_e d\mathbf{v} \equiv \delta T_{\parallel e} n_{0e}$$

Electron momentum eq.

以下の式の各項の低次モーメントを取る.

$$\begin{aligned} \frac{\partial h_e}{\partial t} + \mathbf{v}_E \cdot \nabla h_e + v_{\parallel} \hat{\mathbf{b}} \cdot \nabla h_e &= \frac{q_e}{T_{0e}} \frac{\partial}{\partial t} (\phi - v_{\parallel} A_{\parallel}) \\ + \frac{1}{B_0} \frac{\partial}{\partial y} (\phi - v_{\parallel} A_{\parallel}) &\left[\frac{(n_{0e})'}{n_{0e}} + \left(\frac{v^2}{v_{the}^2} - \frac{3}{2} \right) \frac{(T_{0e})'}{T_{0e}} \right] f_{0e} + \left(\frac{\partial h_e}{\partial t} \right)_c \end{aligned}$$

0th Moment -> Density continuity eq.

1st Moment -> Momentum eq.

Electron momentum eq.

左辺第1項 : $\int v_{\parallel} \frac{\partial h_e}{\partial t} d\mathbf{v} = \frac{\partial}{\partial t} n_{0e} u_{\parallel e}$

左辺第2項 : $\int v_{\parallel} \mathbf{v}_E \cdot \nabla h_e d\mathbf{v} = \mathbf{v}_E \cdot \nabla (n_{0e} u_{\parallel e})$

左辺第3項 : $\int v_{\parallel} v_{\parallel} \hat{\mathbf{b}} \cdot \nabla h_e d\mathbf{v} = \frac{n_{0e}}{2} v_{\text{the}}^2 \hat{\mathbf{b}} \cdot \nabla \left(\frac{q_e \phi}{T_{0e}} + \frac{\delta n_e}{n_{0e}} + \frac{\delta T_e}{T_{0e}} \right)$

右辺第1項 : $\int v_{\parallel} \frac{q_e}{T_{0e}} \frac{\partial}{\partial t} (\phi - v_{\parallel} A_{\parallel}) d\mathbf{v} = -\frac{q_e}{T_{0e}} \frac{\partial}{\partial t} A_{\parallel} \frac{v_{\text{the}}^2}{2} n_{0e}$

Electron momentum eq.

右辺第2項：

$$\int v_{\parallel} \frac{1}{B_0} \frac{\partial}{\partial y} (\phi - v_{\parallel} A_{\parallel}) \left[\frac{(n_{0e})'}{n_{0e}} + \left(\frac{v^2}{v_{\text{the}}^2} - \frac{3}{2} \right) \frac{(T_{0e})'}{T_{0e}} \right] f_{0e} d\mathbf{v}$$
$$= \frac{1}{B_0} \frac{\partial}{\partial y} A_{\parallel} (L_{n_{0e}}^{-1} + L_{T_{0e}}^{-1}) \times \frac{1}{2} n_{0e} v_{\text{the}}^2$$

右辺第3項：

$$\int v_{\parallel} \left(\frac{\partial h_e}{\partial t} \right)_c d\mathbf{v} = \nu_{ei} n_{0e} (u_{\parallel i} - u_{\parallel e})$$

衝突項は， Lenard-Bernstein model

$$\left(\frac{\partial h_e}{\partial t} \right)_c = \nu_{ei} \left[\frac{1}{2} \frac{\partial}{\partial \hat{v}_{\parallel}} \left(\frac{\partial}{\partial \hat{v}_{\parallel}} + 2\hat{v}_{\parallel} \right) h_e + \frac{2\hat{v}_{\parallel} u_{\parallel i}}{v_{\text{the}}} f_{0e} \right. \\ \left. + (1 - 2\hat{v}_{\parallel}^2) f_{0e} \frac{1}{n_{0e}} \int (1 - 2v_{\parallel}'^2) h_e d\mathbf{v}' \right], \quad \hat{v}_{\parallel} = v_{\parallel} / v_{\text{the}}$$

Electron momentum eq.

各項の1次モーメントを整理する.

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \mathbf{v}_E \cdot \nabla$$

$$\begin{aligned} \frac{d}{dt} \left(u_{\parallel e} + \frac{q_e}{m_e} A_{\parallel} \right) = & -\frac{q_e}{m_e} \frac{\partial \phi}{\partial z} - \frac{T_{0e}}{m_e} \hat{\mathbf{b}} \cdot \nabla \left[\frac{\delta n_e}{n_{0e}} + \frac{\delta T_{\parallel e}}{T_{0e}} \right] \\ & + \frac{T_{0e}}{m_e} (1 + \eta_e) \frac{\hat{\mathbf{b}} \cdot \nabla n_{0e}}{n_{0e}} + \nu_{ei} (u_{\parallel i} - u_{\parallel e}) \end{aligned}$$

$$\eta_e = L_{T_{0e}}^{-1} / L_{n_{0e}}^{-1}$$

アンペールの法則より,

$$-\nabla_{\perp}^2 A_{\parallel} = \mu_0 (n_{0e} q_e u_{\parallel e} + n_{0i} q_i u_{\parallel i}) = \mu_0 (n_{0e} q_e u_{\parallel e}) \quad , \because u_{\parallel e} \gg u_{\parallel i}$$

quasi - neutrality

よって,

$$\begin{aligned} \frac{d}{dt} (A_{\parallel} - d_e^2 \nabla_{\perp}^2 A_{\parallel}) = & -\frac{\partial \phi}{\partial z} - \frac{T_{0e}}{q_e} \hat{\mathbf{b}} \cdot \nabla \left[\frac{\delta n_e}{n_{0e}} + \frac{\delta T_{\parallel e}}{T_{0e}} \right] \\ & + \frac{T_{0e}}{q_e} (1 + \eta_e) \frac{\hat{\mathbf{b}} \cdot \nabla n_{0e}}{n_{0e}} + \nu_{ei} d_e^2 \nabla_{\perp}^2 A_{\parallel} \end{aligned} \quad (1)$$

Problem setup

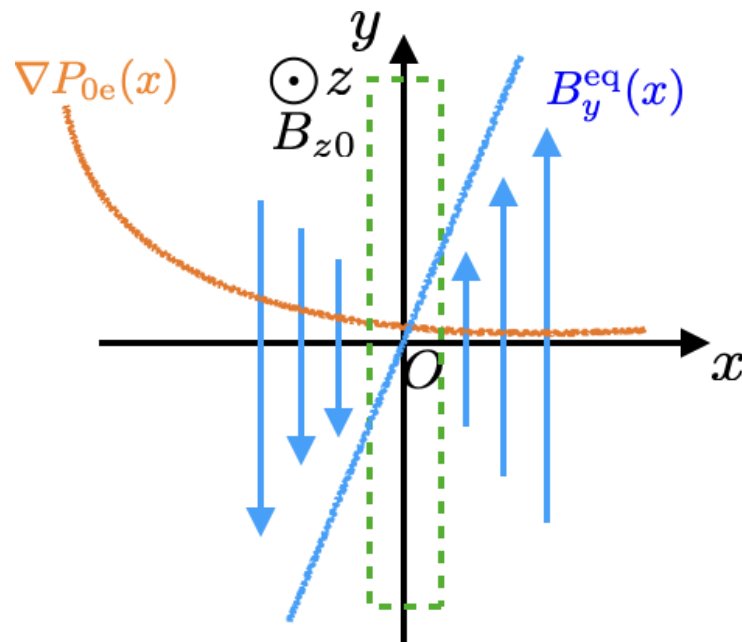
□ 2D slab model

□ 背景平衡磁場

$$\mathbf{B}_0 = B_{z0}\hat{z} + B_y^{\text{eq}}(x)\hat{y}, \quad B_{z0} \gg B_y^{\text{eq}}(x)$$

□ 背景電子压力勾配

$$\begin{aligned} -\nabla P_{0e}(x) &= -\nabla(n_{0e}T_{0e}) \\ &= n_{0e}T_{0e} \left(-\frac{\nabla n_{0e}}{n_{0e}} - \frac{\nabla T_{0e}}{T_{0e}} \right) \\ &= P_{0e} \left(L_{n_{0e}}^{-1} + L_{T_{0e}}^{-1} \right) \hat{x} \end{aligned}$$



Linearization

物理量を平衡量と1次の微小摂動量に分ける.

$$\xi = \xi^{(0)} + \xi^{(1)}, \quad \xi^{(0)} \gg \xi^{(1)}$$

平衡量

$$\begin{aligned} \phi^{(0)} = \delta n_e^{(0)} = \delta T_{\parallel e}^{(0)} = 0 \quad -\frac{d}{dt} A_{\parallel}^{(0)} &= B_y^{\text{eq}(0)} \\ u_{\parallel e}^{(0)} = -\frac{q_e}{m_e} d_e^2 \nabla_{\perp}^2 A_{\parallel}^{(0)} &= \mu_0 \frac{q_e}{m_e} d_e^2 j_{\parallel}^{(0)} \end{aligned}$$

摂動量

$$\xi^{(1)}(x, y, z, t) = \Sigma \tilde{\xi}(x) e^{i(k_y y - \omega t) + i k_z z}$$

Linearization

Eq.(1)に対して線形化を行い整理する.

$$\begin{aligned} i\omega \tilde{A}_{\parallel} - ik_y \frac{B_y^{\text{eq}(0)}}{B_0} \tilde{\phi} &= i(\omega_{*n_{0e}} + \omega_{*T_{0e}}) \tilde{A}_{\parallel} - i\omega d_e^2 \tilde{j}_{\parallel} + \eta \tilde{j}_{\parallel} \\ &\quad + i \frac{T_{0e}}{q_e} k_y \frac{B_y^{\text{eq}(0)}}{B_0} \left(\frac{\delta \tilde{n}_e}{n_{0e}} + \frac{\delta \tilde{T}_{\parallel e}}{T_{0e}} \right) \end{aligned}$$

$$\text{ここで, } \omega_{*n_{0e}, T_{0e}} = \frac{k_y \rho_e v_{\text{the}}}{2L_{n_{0e}, T_{0e}}}, \partial_z = 0$$

$$E_{\parallel} = -\frac{\partial}{\partial t} A_{\parallel} - \nabla_{\parallel} \phi \quad \text{より上式左辺は, } \tilde{E}_{\parallel} = i\omega \tilde{A}_{\parallel} - ik_y \frac{B_y^{\text{eq}(0)}}{B_0} \tilde{\phi} \quad (2)$$

$$\therefore \tilde{E}_{\parallel} = i(\omega_{*n_{0e}} + \omega_{*T_{0e}}) \tilde{A}_{\parallel} - i\omega d_e^2 \tilde{j}_{\parallel} + \eta \tilde{j}_{\parallel} + i \frac{T_{0e}}{q_e} k_y \frac{B_y^{\text{eq}(0)}}{B_0} \left(\frac{\delta \tilde{n}_e}{n_{0e}} + \frac{\delta \tilde{T}_{\parallel e}}{T_{0e}} \right) \quad (3)$$

AstroGK normalization

参照プラズマパラメータ : $\rho_0, L_0, q_0, T_{00}, v_{th0}$

ジャイロ運動論展開パラメータ : $\epsilon = \frac{\rho_0}{L_0} = \frac{k_{\parallel}}{k_{\perp}}$

場の規格化 :

$$A_{\parallel} = \hat{A}_{\parallel} \epsilon \frac{T_{00}}{q_0} \frac{1}{v_{th0}}, \quad \phi = \hat{\phi}_{\parallel} \epsilon \frac{T_{00}}{q_0},$$

$$B_y^{eq(0)} = \hat{B}_y^{eq(0)} \epsilon \frac{B_0}{2}, \quad E_{\parallel} = \hat{E}_{\parallel} \epsilon \frac{T_{00}}{q_0} \frac{1}{L_0}$$

($\hat{}$: normalized params.)

Normalized recon. ele. field

Eq.(2), (3)をAGK norm.を用いて規格化.

$$\text{Eq.(2): } \hat{E}_{\parallel} = i\hat{\omega} \hat{A}_{\parallel} - i\hat{k}_y \frac{\hat{B}_y^{\text{eq}(0)}}{2} \hat{\phi} \quad (4)$$

$$\begin{aligned} \text{Eq.(3): } \hat{E}_{\parallel}^{\text{ohm}} = & i\hat{\omega}_{*e} \hat{A}_{\parallel} - i\hat{\omega} \frac{2\hat{m}_e}{n_{0e}\hat{q}_e} \hat{j}_{\parallel} \\ & + 0.38\hat{\nu}_{ei} \frac{2\hat{m}_e}{n_{0e}\hat{q}_e} \hat{j}_{\parallel} + i\hat{k}_y \frac{\hat{B}_y^{\text{eq}(0)}}{2} \frac{\hat{T}_{0e}}{\hat{q}_e} \left(\delta\hat{n}_e + \delta\hat{T}_{\parallel e} \right) \end{aligned} \quad (5)$$

※AGKでは, Spitzer resistivityなので
係数が0.38.

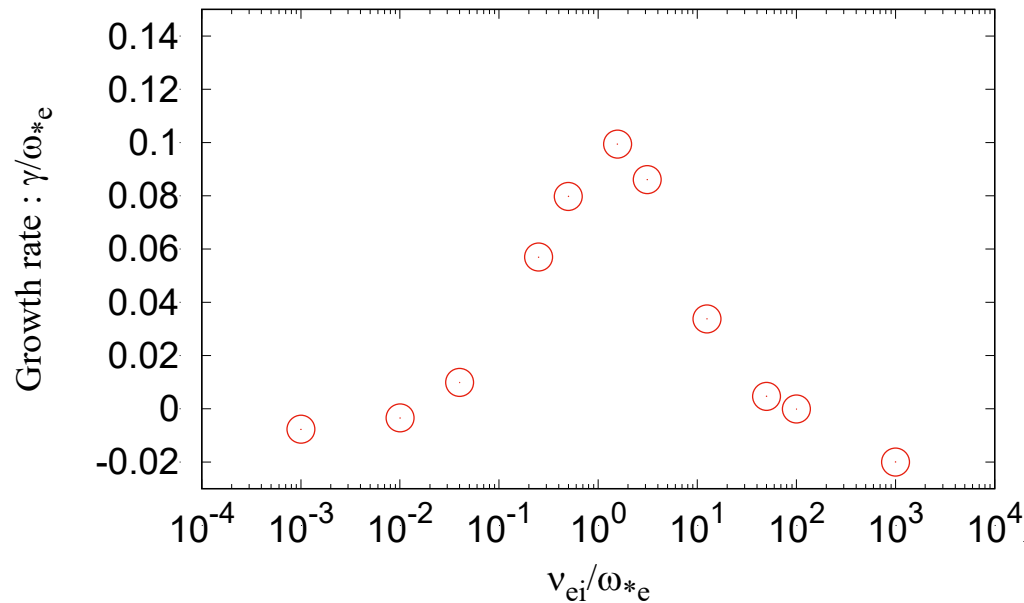
Verification

Eq.(4) と (5) の計算結果の比較を行う.

$$\beta_e = 0.024, \tau = T_{0i}/T_{0e} = 1$$

$$n_{0e} = n_{0i} = 1, \eta_e = 1$$

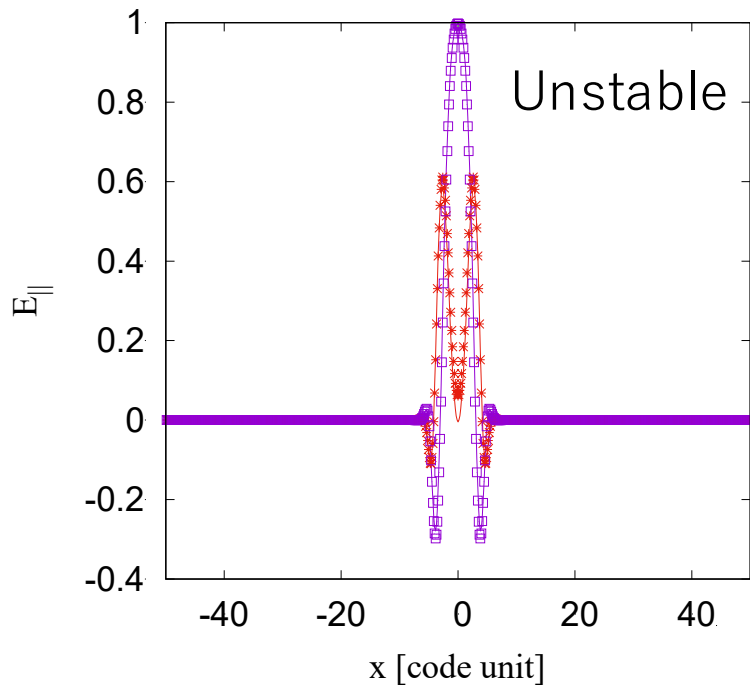
$$\Delta'a = -4.58, k_y \rho_i = 0.05, \eta_e = 1$$



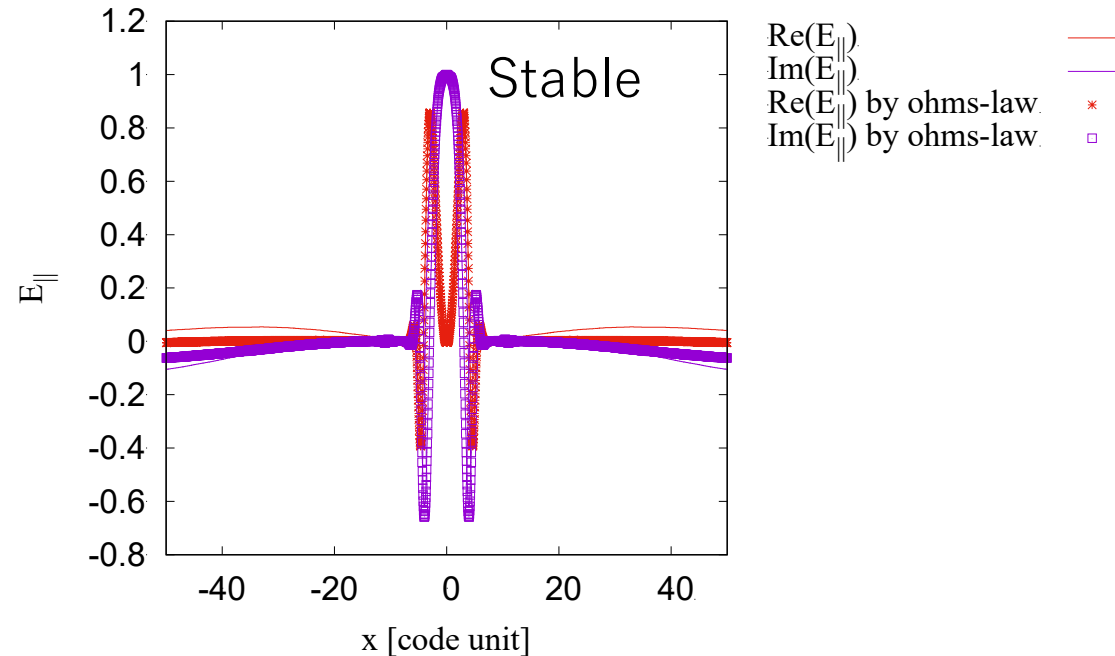
Verification

Eq.(5)と(6)の計算結果は概ね一致している.

$$\nu_{ei}/\omega_{*e} = 0.25$$

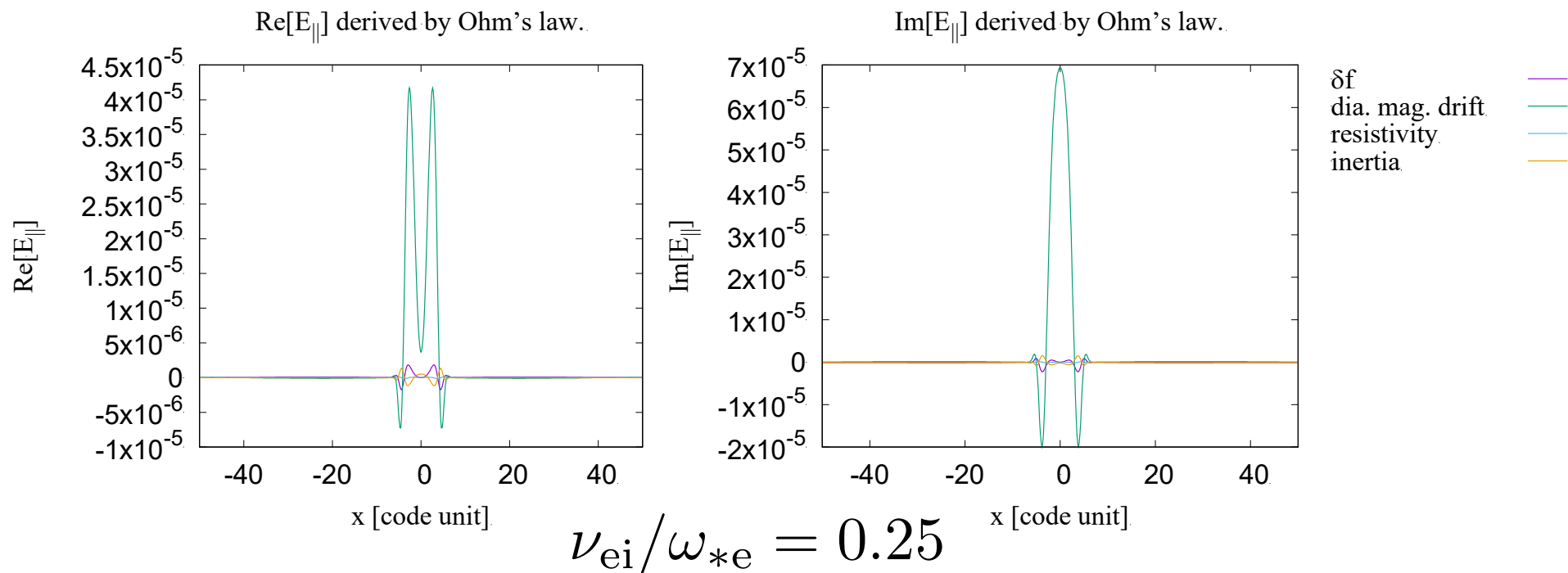


$$\nu_{ei}/\omega_{*e} = 0.01$$



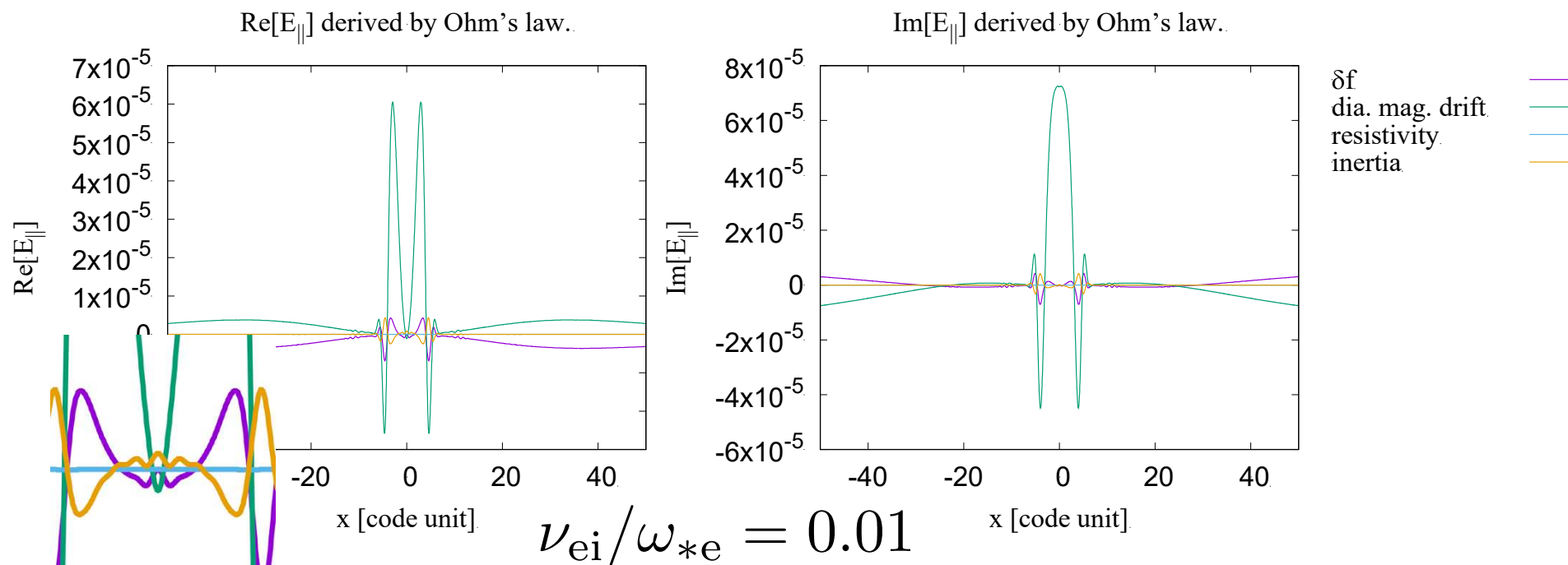
Components of Eq.(5) for Unstable

反磁性ドリフト項が支配的.
衝突の効果は非常に小さい.



Components of Eq.(5) for stable

x=0において反磁性ドリフト項と電子慣性項がバランスして電場~0になる.



Summary

- 電子の運動論効果を内包したOhm's lawを導出し，Faraday's lawより導出されるリコネクション電場と比較.
- 概ね一致するが，実部が一致しない.
- 電場形成には，反磁性ドリフト項が支配的.
- 無衝突の領域では電子慣性効果と反磁性効果がバランスし電場が0になる？

Tasks

- 実部の不一致の原因について調査. 反磁性ドリフトが支配的なら一致してほしい.
- MT不安定性の衝突周波数スキャンを再度実施. 出力データがないため.

以上です.



Ordering in Zocco+, 2014

Low-beta $\beta_e \sim \frac{m_e}{m_i} \ll 1$

$$\frac{d}{dt} \sim \omega \sim k_{\perp} u_{\perp} \sim k_{\perp}^2 \frac{q_e \phi}{B_0}$$

$$\frac{Z}{\tau} \sim 1 \text{ のとき } \frac{d}{dt} \frac{\delta n_e}{n_{0e}} \sim \frac{q_e \phi}{T_{0e}} \omega \sim \frac{\epsilon}{\sqrt{\beta_e}} \omega$$

$$u_{\parallel e} \sim \frac{\epsilon}{\sqrt{\beta_e}} v_{\text{the}} \text{ より}$$

$$\hat{\mathbf{b}} \cdot \nabla u_{\parallel e} \sim \epsilon k_{\perp} \frac{\epsilon}{\sqrt{\beta_e}} v_{\text{the}} \sim \frac{\epsilon}{\sqrt{\beta_e}} \omega$$



Electron kinetic eq.

$$\frac{\partial h_e}{\partial t} + \mathbf{v}_E \cdot \nabla h_e + v_{\parallel} \hat{\mathbf{b}} \cdot \nabla h_e = q_e \frac{f_{0e}}{T_{0e}} \frac{\partial}{\partial t} (\phi - v_{\parallel} A_{\parallel}) - \frac{1}{B_0} \mathbf{e}_z \cdot \nabla (\phi - v_{\parallel} A_{\parallel}) \times \nabla f_{0e} + \left(\frac{\partial h_e}{\partial t} \right)_c$$

$$\text{ここで, } \mathbf{v}_E \cdot \nabla = \frac{1}{B_0} \{ \phi, \}^{\wedge}, \quad \hat{\mathbf{b}} \cdot \nabla = \partial_z - \frac{1}{B_0} \{ A_{\parallel}, \}^{\wedge}$$

$$\text{Poisson括弧: } \{a, b\} = \frac{\partial a}{\partial x} \frac{\partial b}{\partial y} - \frac{\partial a}{\partial y} \frac{\partial b}{\partial x}$$

$$\text{平衡Maxwellian: } f_{0e}(x) = \frac{n_{0e}(x)}{(\sqrt{\pi} v_{\text{the}}(x))^3} e^{-v^2/v_{\text{the}}^2(x)}$$

$$\text{衝突項: } \left(\frac{\partial h_e}{\partial t} \right)_c$$

Electric field

一般的に電場 $\mathbf{E}$ は以下で与えられる.

$$\mathbf{E} = -\frac{\partial}{\partial t}\mathbf{A} - \nabla\phi \quad (\text{By Faraday's law})$$

リコネクション電場は背景磁場 $\mathbf{B}_0$ に対して平行な成分なので以下で求まる.

$$\hat{\mathbf{b}} \cdot \mathbf{E} = -\frac{\partial}{\partial t}\hat{\mathbf{b}} \cdot \mathbf{A} - \hat{\mathbf{b}} \cdot \nabla\phi, \quad \hat{\mathbf{b}} = \mathbf{B}_0/B_0$$

$$\therefore E_{\parallel} = -\frac{\partial}{\partial t}A_{\parallel} - \nabla_{\parallel}\phi \quad (1)$$



Diamagnetic Drift Frequency of Electron

- ドリフト速度

$$\mathbf{U}^d = -\frac{\nabla P_0 \times \mathbf{B}_0}{qn|\mathbf{B}_0|^2}$$

$$\nabla P_0 = -P_0(L_{n0}^{-1} + L_{T0}^{-1})\hat{x} \quad \mathbf{B}_0 = B_{z0}\hat{z}$$

y軸正方向への電子ドリフト速度

$$U_{n_{0e}, T_{0e}}^d = \frac{T_{0e}}{eB_{z0}L_{n_{0e}, T_{0e}}} \quad (\because q = -e)$$

- 電子のドリフト周波数

$$\omega_{n_{0e}, T_{0e}}^* = k_y U_{n_{0e}, T_{0e}}^d$$

$$\omega_e^* = \omega_{n_{0e}}^* + \omega_{T_{0e}}^*$$